KARTHIK CHANDRAN LAKSHMANASAMY

Alexandria, VA | U.S. Permanent Resident | No Sponsorship Required
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PROFESSIONAL SUMMARY

Principal Software Engineer in Test with 9+ years of experience leading quality engineering across SaaS, cloud, distributed, data-intensive, and AI/LLM platforms. Expertise in Python/Pytest automation, UI/API testing, CI/CD, performance and reliability engineering, cloud-native testing, and risk-based test strategy. Modernized legacy test frameworks, validated RAG/LLM and semantic layer systems, built automation across multi-cloud environments, and mentored engineers.

CORE EXPERTISE

AI/LLM and RAG Testing, Agentic AI and MCP Testing, Python, Pytest, Selenium, Playwright, WebdriverIO, API and Integration Testing, Performance and Concurrency Testing, CI/CD, Docker, Kubernetes, AWS, Azure, GCP, Distributed Systems, Framework Architecture and Modernization

PROFESSIONAL EXPERIENCE

Principal Software Engineer in Test **May 2023 - Present**
MicroStrategy *Tysons, VA*

- Modernized a 20-year-old legacy ETL, SQL, and data-validation automation framework by migrating core regression workflows to Python/Pytest and containerizing execution with Docker/Kubernetes.
- Developed end-to-end automation using WebdriverIO and Python for Strategy Mosaic, expanding regression coverage across enterprise analytics and data workflows.
- Built automated validation for Mosaic Universal Semantic Layer, including semantic models, metrics, relationships, ETL transformations, and SQL accuracy across enterprise data sources and BI platforms.
- Automated AI-driven data modeling workflows, including linked models, base and derived metrics, automatic metric generation, and AI-powered analytics.
- Developed automated MCP integration testing with Claude for connector setup, tool discovery, semantic-model access, query execution, accuracy, permissions, and execution modes.
- Built an LLM-as-a-Judge framework evaluating AI responses against a governed semantic layer for grounding, consistency, hallucination risk, accuracy, and semantic correctness.
- Implemented predictive and RAG testing using Python and TensorFlow for conversational AI and forecasting responses.
- Built Python automation for cloud deployments across AWS, Azure, GCP, OpenShift, and StackIT, validating provisioning, health, upgrades, autoscaling, concurrency, recovery, and stability.
- Automated pre/post-deployment validation using AWS CodeBuild and designed Jenkins CI/CD workflows for functional, integration, and regression testing.
- Built Python concurrency and performance-testing frameworks for event-driven data models to identify throughput, latency, and scalability bottlenecks.
- Mentored junior engineers on automation architecture, cloud testing, and tooling selection, helping achieve 90% automated test coverage.

Senior Software Engineer in Test **Jul 2019 - May 2023**
MicroStrategy *Poland*

- Designed an automated framework using Protractor, JavaScript, and Cucumber for SAP S/4 Connector and cloud-based applications and services.
- Improved Kafka-Flink data-stream performance by 38% through batch-size and processing-interval optimization and validation under data-intensive workloads.
- Contributed to certification and release validation for ADLS, S3, and GCS cloud connectors in MicroStrategy 2021.
- Led FedRAMP certification testing for the Cloudera Kubernetes platform against federal security and platform requirements.
- Built a data pipeline to analyze server logs and surface performance metrics to optimize backend services.

Software Engineer in Test **Mar 2018 - Jul 2019**
MicroStrategy *Tysons, VA*

- Designed a Pytest and Selenium framework to detect Google BigQuery data-import issues and improve data connectivity reliability.
- Integrated automated tests into CI/CD pipelines with DevOps, improving continuous release validation.
- Managed planning, integration, and continuous testing for the Google BigQuery connector.
- Led Teradata-to-Google BigQuery migration validation with cross-functional teams.

Associate Software Engineer in Test **Mar 2017 - Mar 2018**
MicroStrategy *Tysons, VA*

- Designed a performance-testing framework benchmarking big-data connectors across large-scale workloads.
- Improved large-scale data-import response time by 44% using Google Storage API and performance-focused validation.
- Led improvements to test efficiency, flaky tests, and defect detection; collaborated on requirements and feature priorities.
- Implemented and validated Kerberos authentication for Cloudera Data Platform using Active Directory.

TECHNICAL SKILLS

Programming: Python, JavaScript, TypeScript, Java, Bash, Groovy, C++

Test Automation: Pytest, Selenium, Playwright, WebdriverIO, Cypress, Protractor, Cucumber, JUnit, TestNG

AI / LLM Testing: Generative AI, LLM Evaluation, LLM-as-a-Judge, RAG Validation, Response Grounding, TensorFlow, Azure Embeddings, LangChain, OpenAI, AWS Bedrock, Gemini, Claude, GPT-OSS, MCP

API and Integration: Postman, Swagger, REST APIs, Microservices, Data Connectors

Performance: JMeter, Gatling, Concurrency Testing, Load Testing, Scalability, Regression Testing

CI/CD and DevOps: Jenkins, AWS CodeBuild, Git, Docker, Kubernetes, Terraform, Ansible

Cloud: AWS, Azure, GCP, OpenShift, StackIT

Data and Distributed Systems: Kafka, Flink, BigQuery, Databricks, Snowflake, Oracle, MySQL, MongoDB, PostgreSQL, Redshift, Redis

BI and Enterprise: MicroStrategy, Tableau, Qlik, Salesforce, ServiceNow, Power BI, ThoughtSpot, Amazon QuickSight

EDUCATION

Master of Science in Computer Science | Texas State University | Aug 2013 - May 2016
Bachelor of Engineering in Computer Science | Anna University | Sep 2009 - May 2013

PUBLICATIONS & PACKAGES

Pawzok - Open-source Python package for testing distributed systems and end-to-end data workflow validation. PyPI | GitHub
Numvo - Open-source Python package for phone-number spam and reputation-risk checks. PyPI | GitHub